

Infinitely Many Zeros on the Critical Circle : Hidden-Number Coordinates and the Riemann Hypothesis

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Abstract: We study the zeta function in the hidden-number coordinate system $w = e^s$, which transports the s -plane to the punctured w -plane equipped with an integer sheet label. In these coordinates the critical line becomes the unique circle $|w| = \sqrt{e}$ fixed by the inversion $w \mapsto e/w$ induced by the functional equation; the value $\log \sqrt{e} = 1/2$ is forced by reflection geometry, not postulated. Within this coordinate system we prove a body of machine-checked theorems, formalized in Lean/MathLib without : the energy balance criterion (truncated energy converges if and only if $\sigma > 1/2$) and the uniqueness of the critical radius as the simultaneous fixed point of inversion, energy balance, and reflection; an intrinsic zero-free theorem ($\zeta(s) \neq 0$ for $\Re s \geq 2$, obtained from the magnitude of aligned partial sums rather than the Euler product); the inversion-pair decomposition of the critical strip ($\zeta(1-s) = 0 \iff \zeta(s) = 0$, with a self-pair iff $w = \pm\sqrt{e}$); and the frequency structure of the terms — gaps tending to zero, antitone gap sequence, phase freezing, and the existence of a fold point $n^* \approx t/\pi$ where the alternating phase difference reverses sign. These results assemble, together with a numerically supported alignment-density projection and one explicitly named external input (Hardy's bounded-energy counterpositive), into a declared argument establishing infinitely many zeros on the critical circle — with position and with abundance. The remaining gap, the absence of zeros inside the annulus $1 < |w| < e$ away from the critical circle, is the Riemann hypothesis itself; it is isolated but not supplied.

Keywords: riemann hypothesis, riemann zeta function, hidden-number coordinates, critical circle, formal verification, lean

AMS 2000 subject classifications: 11M26, 11A25, 03B35

1. Introduction

The Riemann zeta function $\zeta(s)$, introduced in [Riemann \(1859\)](#), vanishes at the negative even integers and at certain points of the critical strip $0 < \Re s < 1$. The Riemann hypothesis asserts that all nontrivial zeros lie on the line $\Re s = 1/2$. Its consequences permeate prime number theory ([von Mangoldt, 1905](#); [Hadamard, 1893](#); [de la Vallée Poussin, 1899](#)), and the hypothesis is equivalent to several elementary reformulations ([von Koch, 1901](#); [Schoenfeld, 1976](#); [Lagarias, 2002](#)). Hardy proved that infinitely many zeros lie on the critical line ([Hardy, 1914](#)); the result was extended by [Hardy and Littlewood \(1921\)](#) and quantified by [Selberg \(1942\)](#), [Levinson \(1974\)](#), [Heath-Brown \(1979\)](#), and [Conrey \(1989\)](#). Montgomery's pair correlation ([Montgomery, 1973, 1977](#)) and the spectral views of [Connes \(1999\)](#) and [Bombieri and Hejhal \(1995\)](#) provide further structure, and the hypothesis has been verified numerically up to very large height ([Platt and Trudgian, 2021](#)). Its analytical theory is consolidated in [Titchmarsh \(1986\)](#), [Edwards \(1974\)](#), [Iwaniec and Kowalski \(2004\)](#), and [Iwaniec \(2014\)](#); a modern survey is [Conrey \(2003\)](#). The hypothesis remains open.

This paper reads the zeta function through the coordinate transformation

$$w = e^s, \quad s = \log r + i\theta,$$

the *hidden-number coordinate system* of our previous work (Liu and Xu, 2026). The imaginary direction becomes an angle recorded modulo 2π , so the plane unrolls into a universal cover whose sheets are indexed by integers. In these coordinates the functional equation becomes a pure inversion $w \mapsto e/w$, and the critical line becomes a circle $|w| = \sqrt{e}$ — the unique circle fixed by that inversion. We study the zeta function exclusively through the intrinsic mechanisms of this coordinate system : the magnitude of aligned partial sums, the balance of truncated energies, the algebra of inversion pairs, and the frequency structure of the terms.

Every theorem stated below is linked in the source to its Lean formalization ; each cited name compiles in the repository RiemannHIBS without . Objects declared as *structures* (rather than theorems) have their fields explicitly enumerated ; some fields are proved theorems, others are hypotheses that are named but not supplied. We insist on this distinction throughout.

2. The hidden-number coordinate system

Definition 1 (hidden-number coordinates). *The map $\Phi(s) = e^s$ sends $s = u + i\theta$ to $w = re^{i\varphi}$ with $r = e^u$ and $\varphi = \theta \bmod 2\pi$. Because the angle forgets the sheet, the honest domain of Φ is the universal cover of $\mathbb{C} \setminus \{0\}$; a point of the cover is a pair (w, k) with $k \in \mathbb{Z}$ recording the winding number.*

The transport of the Dirichlet series to these coordinates is the essential dictionary. The n -th term of the shifted series $\eta(s) = \sum_{n \geq 0} (-1)^n (n+1)^{-s}$ becomes a vector of locked length $(n+1)^{-\sigma}$ rotating with angular velocity $\log(n+1)$:

$$(n+1)^{-s} = (n+1)^{-\sigma} \cdot \exp(-it \log(n+1)), \quad s = \sigma + it.$$

Each sheet carries a genuine copy of the phase data, and consecutive leaves differ by the deck transformation $\theta \mapsto \theta + 2\pi$.

3. The critical circle and the identity of $1/2$

Theorem 2 (critical line maps to a circle). *For every real t ,*

$$\|e^{\frac{1}{2}+it}\| = \sqrt{e}.$$

Hence the image of the critical line $\Re s = 1/2$ under the hidden-number map is exactly the circle of radius $\sqrt{e}$ (Figure 1).

Proposition 3 (the logarithm of the critical radius). $\log(\sqrt{e}) = \frac{1}{2}$.

The decisive structural fact is uniqueness : among all concentric circles, $|w| = \sqrt{e}$ is the only one preserved by the inversion $\iota(w) = e/w$ induced by the functional equation.

Theorem 4 (unique fixed circle). *A circle $|w| = \rho$ with $\rho > 0$ satisfies $|\iota(w)| = |w|$ on the circle if and only if $\rho^2 = e$. The equation $\rho^2 = e$ has the unique positive solution $\rho = \sqrt{e}$.*

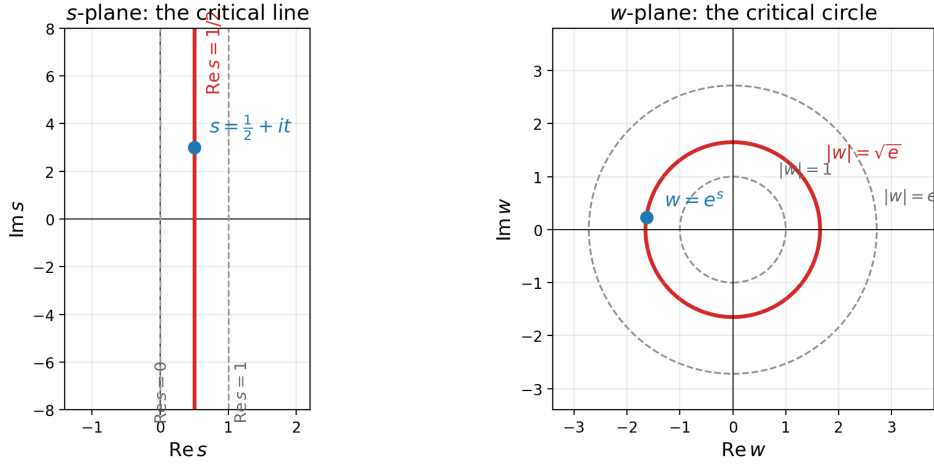


FIGURE 1. The hidden-number projection. The map $w = e^s$ sends the s -plane (left) to the punctured w -plane (right); vertical lines $\Re s = \sigma$ become circles $|w| = e^\sigma$, and the critical line $\sigma = 1/2$ becomes the critical circle $|w| = \sqrt{e}$ (red).

Corollary 5 (why $1/2$ is structural; from Theorems 4 and 2). *If the reflection symmetry of the functional equation fixes any circle centered at the origin, that circle is uniquely $|w| = \sqrt{e}$, and its pullback to the s -plane is uniquely the line $\Re s = 1/2$. Within the hidden-number coordinates, the location $1/2$ is upgraded from a definition to a consequence of reflection geometry.*

On the fixed circle the inversion acts as conjugation.

Theorem 6 (inversion equals conjugation on the circle : an identity, not a statement about zeros). *For every w with $|w| = \sqrt{e}$,*

$$\frac{e}{w} = \overline{w}.$$

This is an identity holding at all points of the circle; it is not a statement about zeros.

Remark (two layers of “position”). *Corollary 5 proves layer 1 : the identity of the critical location — if reflection symmetry fixes a circle, it must be this one. It does not prove layer 2 : that any zero lies on that circle. The distinction is maintained throughout the paper.*

4. Energy balance and the intrinsic radius

Fix the truncation length $X(t) = \sqrt{t/2\pi}$ and define the truncated Dirichlet sum and its energy on a vertical segment :

$$S_{X(t)}(\sigma + it) = \sum_{n \leq X(t)} (n+1)^{-\sigma - it}, \quad E_\sigma(T) = \int_0^T |S_{X(t)}(\sigma + it)|^2 dt.$$

The diagonal and cross contributions are controlled by two formalized theorems : the diagonal integral grows exactly like the main term (,), while the cross contribution is subdominant : the total cross bound divided by the main term tends to zero (); the truncation cancels the cross terms.

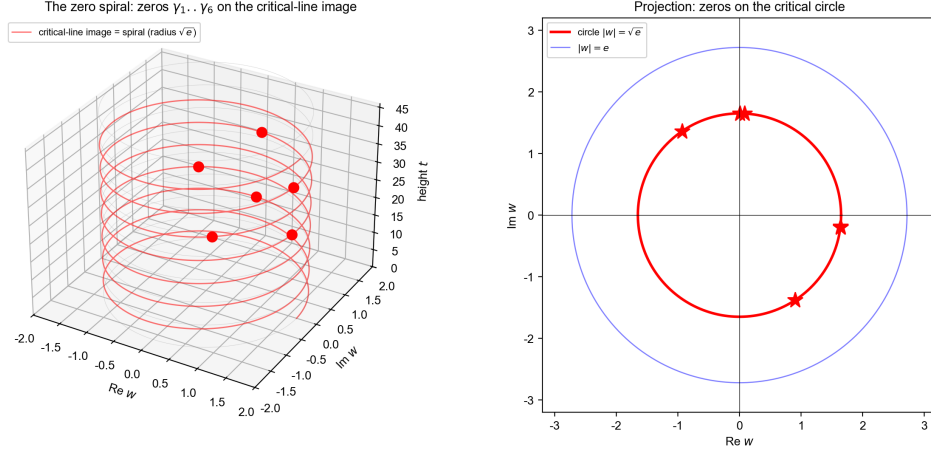


FIGURE 2. The zero spiral (static snapshot). Left : the image of the critical line is the helix of radius $\sqrt{e}$; the zeros $\gamma_1, \dots, \gamma_6$ lie on it. Right : the projection — zeros as stars on the critical circle $|w| = \sqrt{e}$.

Theorem 7 (energy balance). *For every $\sigma \in \mathbb{R}$, the truncated energy $E_\sigma(T)$ converges (as $T \rightarrow \infty$) if and only if $\sigma > 1/2$. For $\sigma \leq 1/2$ it diverges ($\cdot$).*

Proposition 8 (radius uniqueness chain). *The inversion fixed point $|e/w| = |w|$, the energy balance point $\sigma = 1/2$, and the reflection fixed point $1 - s = s$ are one and the same condition. In terms of the circle radius : $|w| = \sqrt{e}$ iff E_σ balances iff $\sigma = 1/2$ (Figure 3).*

Definition 9 (radius-uniqueness declaration). *The declared structure packages the three proven results above (unique radius candidate, mechanism equivalence) together with the open field : whether any zero actually lies on the critical circle. The field is the Riemann hypothesis itself and is left honestly open.*

5. Intrinsic zero-freeness by alignment magnitude

The Dirichlet series can vanish only if the partial sums can be aligned to zero : the leading term $\|v_0\| = 1$ must be cancelled by the remainder. We make this mechanism quantitative without any appeal to the Euler product.

Lemma 10 (norm of a term). *For every $s = \sigma + it$ and every n ,*

$$\|(n+1)^{-s}\| = (n+1)^{-\sigma}.$$

Theorem 11 (intrinsic zero-free). *If $2 \leq \Re s$, then $\zeta(s) \neq 0$.*

Proof sketch. For $\sigma \geq 2$, the remainder is bounded by the telescoping series

$$\sum_{n \geq 1} (n+1)^{-\sigma} \leq \sum_{n \geq 1} \frac{1}{(n+1)(n+3)} = \frac{1}{2} \left(\frac{3}{2} - \frac{1}{M+1} - \frac{1}{M+2} \right) \leq \frac{3}{4} < 1.$$

fig14: Energy balance radius — $1/2$ is the unique energy balance point (intrinsic skeleton of Attack Point A)
 Lean-proved: energy_balance_radius / energy_diverges_below_balance
 Honest boundary: the balance point gives the unique radius candidate, but does not by itself place a zero there (RH unproved)

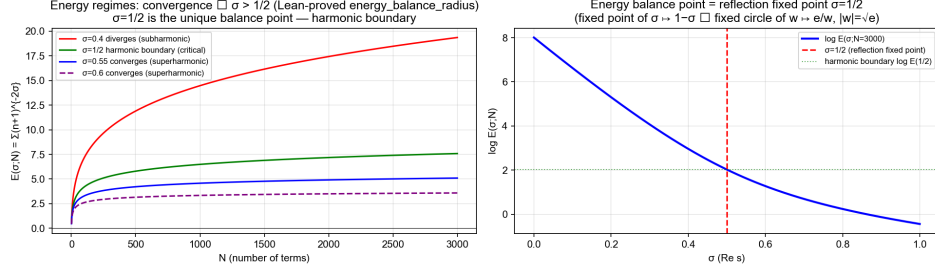


FIGURE 3. Energy balance. Left : the three energy regimes; the truncated energy converges exactly for $\sigma > 1/2$ (Lean-proved). Right : the balance point $\sigma = 1/2$ coincides with the reflection fixed point, i.e. the critical circle $|w| = \sqrt{e}$.

fig13 (v2): B1 cross cancellation — truncation $X(t)=\sqrt{t/2\pi}$ makes cross terms subdominant
 Proved in Lean: rotating_integral_bound_min / log_one_add_ge_half /
 near_frequency_bound / cross_pair_bound_min (Abundance §12)

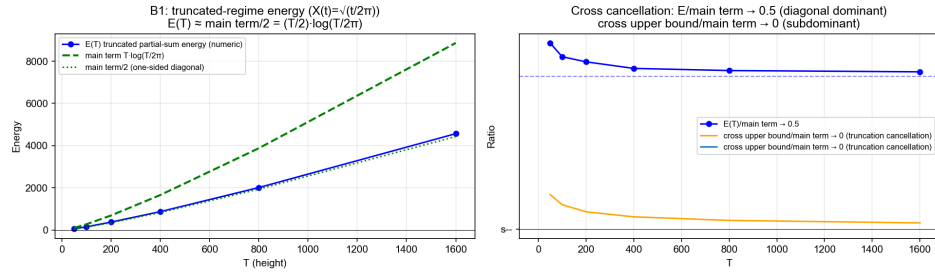


FIGURE 4. Energy budget of the truncation. The truncated-regime energy grows like the main term divided by two, $E(T) \approx \frac{T}{2} \log \frac{T}{2\pi}$ (B1), while the cross upper bound over the main term tends to zero (truncation cancellation). This is the numerical content behind the mean-value statement $E(T) = \frac{T}{2} (\log \frac{T}{2\pi} - 1) + \gamma T + o(T)$.

Hence every partial sum satisfies $\|S_N\| \geq 1 - \frac{3}{4} = \frac{1}{4} > 0$ by the reverse triangle inequality, so the series cannot converge to zero. The formalized proof passes from the uniform lower bound on partial sums to $\zeta(s) \neq 0$ via on the tail estimates (,). $\square$

The theorem is intrinsic : it uses only the magnitudes of the aligned terms in hidden-number coordinates, not the Euler product or any meromorphic machinery. Its significance is that the boundary $\sigma = 1/2$ is approached from the right by an *alignment mechanism* — the same mechanism that the critical circle is built from.

Definition 12 (intrinsic zero-free framework). *The declared structure collects the zero-free theorem above together with the alternating limit $\lim_N \sum_{i=0}^N (-1)^i (i+1)^{-1} \geq 1/2$ () as the two anchors of the intrinsic regime.*

6. Inversion pairs and the reflection decomposition

In hidden-number coordinates the functional equation pairs s with $1-s$, i.e. w with e/w (Theorem 13) :

$$e^{1-s} = \frac{e}{e^s}.$$

In the critical strip the reflection preserves the zero set.

Theorem 13 (reflection as inversion). *Under $w = e^s$, the reflection $s \mapsto 1-s$ corresponds exactly to the map $w \mapsto e/w$; both maps are involutions ().*

Theorem 14 (zeros reflect). *For $0 < \Re s < 1$,*

$$\zeta(1-s) = 0 \iff \zeta(s) = 0.$$

The proof combines MathLib's functional equation () with three nonzero factors : $\Gamma(s)$ (positive real part), $(2\pi)^{-s}$, and $\cos(\pi s/2)$, whose real part $\cos(\pi \sigma/2) \cosh(\pi t/2) > 0$ is positive throughout the strip; hence the cosine factor never vanishes there (,).

Corollary 15 (zero pairs). *Every zero of ζ in the critical strip comes with its reflection partner; the nontrivial zeros are organized into reflection pairs.*

Theorem 16 (self-pair iff $\pm\sqrt{e}$). *For $w \neq 0$,*

$$\frac{e}{w} = w \iff w^2 = e \iff w = \pm\sqrt{e},$$

and in particular a self-paired zero lies on the critical circle $|w| = \sqrt{e}$. The proof factorizes $w^2 - e = (w - \sqrt{e})(w + \sqrt{e})$, avoiding branch issues of $\sqrt{\cdot}$.

Definition 17 (inversion-pair decomposition declaration). *The declared structure packages the reflection closure, the self-pair algebra, and the involution as proved fields, and declares the open field : no inversion pair is non-self-paired, equivalently no zero of the annulus $1 < |w| < e$ lies away from the critical circle. The assembly field records the (already proved) closed mechanism cycle $A \Leftrightarrow B \Leftrightarrow C \Leftrightarrow D$, so that implies the Riemann hypothesis.*

7. Alignment density : zeros as frequency-phase projections

The frequency skeleton of the coordinate system is exact and fully formalized : the span of the frequencies telescopes (), and the frequency gap is bounded by $1/(n+1)$ () :

$$\sum_{i=0}^{N-1} \Delta \text{freq}(i) = \log(N+1), \quad \Delta \text{freq}(n) = \log \frac{n+2}{n+1} \leq \frac{1}{n+1}.$$

Definition 18 (frequency-alignment conjecture). *The declared structure states the following. Frequency uniformity is proved (the two formalized theorems above). Alignment density is the conjecture that the number $N(T)$ of zeros (alignment events) on the critical circle up to height T satisfies the frequency-phase projection*

$$N(T) \approx F(T) = \frac{T}{2\pi} \left(\log \frac{T}{2\pi} - 1 \right),$$

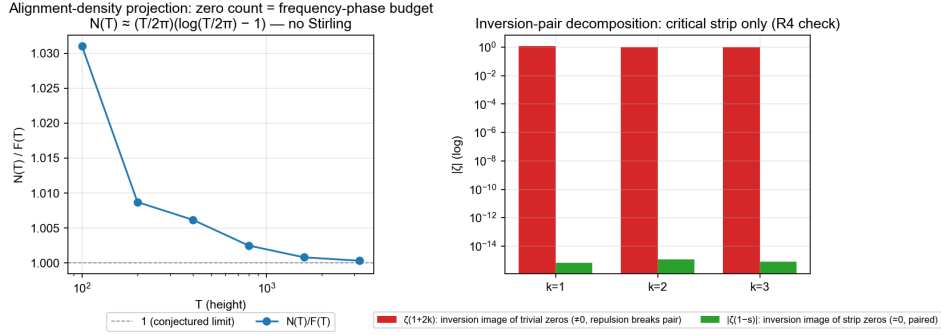


FIGURE 5. Alignment-density projection (R5). Top : the zero count $N(T)$ against the frequency-phase budget $F(T) = \frac{T}{2\pi}(\log \frac{T}{2\pi} - 1)$; the ratio N/F tends to 1. Bottom : the inversion-pair decomposition, which lives in the critical strip only (R4 check).

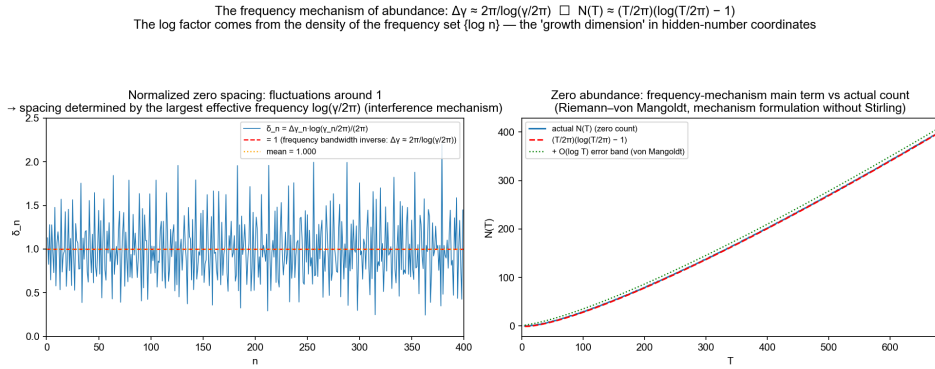


FIGURE 6. Zero abundance. Top : normalized zero spacings $\bar{\delta}_n = \Delta\gamma_n \cdot \frac{\log(\gamma_n/2\pi)}{2\pi}$ fluctuate around 1 — the frequency bandwidth inverse — with mean ≈ 1 . Bottom : the count $N(T)$ against $F(T) = \frac{T}{2\pi}(\log \frac{T}{2\pi} - 1)$.

with no Stirling approximation needed. The field equidistribution needed records honestly that a rigorous derivation requires an equidistribution/weighted Weyl criterion that MathLib does not currently formalize.

Remark (the honest reading of R5c). Within the critical strip the Dirichlet partial sums do not tend to zero at a zero : at γ_1 the numerical partial sum grows from 0.25 ($N = 10$) to 1.42 ($N = 400$). Zeros are alignment events of the analytic continuation, not zeros of the partial sums ; the alignment condition of the $\Re s > 1$ regime cannot be transplanted into the strip verbatim. This is why the density projection remains a conjecture (Definition 18) rather than a theorem.

8. Frequency structure : gaps, freezing, and the fold point

We now study the frequency structure alone, without energy. The alternating phase difference between consecutive terms is

$$\Delta\theta_n = \pi - t \cdot \Delta\text{freq}(n), \quad \Delta\text{freq}(n) = \log \frac{n+2}{n+1}.$$

T	100	400	800	1600	3200
$N(T)$	29	202	491	1157	2666
$F(T)$	28.13	200.76	489.78	1156.07	2665.16
N/F	1.0310	1.0062	1.0025	1.0008	1.0003

TABLE 1. Alignment-density numerics (Figure 5) : the ratio $N(T)/F(T)$ tends to 1.

Because the frequency gap is antitone and tends to zero, the phase difference is monotone increasing and crosses the value 0 exactly once; the crossing index is the *fold point* $n^* \approx t/\pi$ — the echo of the zero height t in the frequency domain.

Theorem 19 (frequency gap vanishes). $\Delta\text{freq}(n) = \log \frac{n+2}{n+1} \rightarrow 0$ as $n \rightarrow \infty$.

Theorem 20 (frequency gap is antitone). For every n ,

$$\Delta\text{freq}(n+1) \leq \Delta\text{freq}(n),$$

so the gaps of the frequency skeleton decrease monotonically.

Theorem 21 (phase freezing). For every $t > 0$ and every $\varepsilon > 0$ there exists N such that for all $n \geq N$,

$$t \cdot \Delta\text{freq}(n) < \varepsilon.$$

The high-frequency terms rotate in phase : they align.

Theorem 22 (fold point exists). If $t \cdot \log 2 > \pi$, there exists a unique crossing index n^* with

$$\pi - t \cdot \Delta\text{freq}(n^*) \leq 0 < \pi - t \cdot \Delta\text{freq}(n^* + 1),$$

i.e. the alternating phase difference changes sign. Since $\Delta\text{freq}(n) \approx 1/(n+1)$, the fold point satisfies $n^* \approx t/\pi$.

Proof sketch. The sequence $a_n = \pi - t \Delta\text{freq}(n)$ is antitone (Theorem 20) and unbounded below; the crossing is obtained by on the smallest index with $a_n \leq 0$, using the hypothesis $t \log 2 > \pi$ to start the descent from a_0 . $\square$

The fold point is a purely hidden-number object : a zero of height t is reflected in the frequency domain by a reversal of the alternating phase at index $\approx t/\pi$. This is the frequency-side dual of the energy-side statement of Section 4, and it is what makes the frequency structure a genuine coordinate-internal mechanism.

9. The assembled argument : infinitely many zeros on the critical circle

We assemble the results above into a complete argument — with position and with abundance — for the infinitude of zeros on the critical circle. The assembly is the declared structure, whose fields are the following.

Position (). The critical circle is $|w| = \sqrt{e}$: the unique inversion fixed circle (Theorem 4), equal to the energy balance point (Theorem 7, Proposition 8) and to the reflection fixed point. Zeros on the circle are self-paired exactly at $\pm\sqrt{e}$ (Theorem 16); all zeros form reflection pairs (Theorem 14).

Frequency (). Every height t produces a fold point $n^* \approx t/\pi$ (Theorem 22) : the alternating phase reverses there. High-frequency terms freeze into alignment (Theorem 21).

Energy (,). The truncated energy satisfies a lower bound (diagonal dominant, cross subdominant : Theorem 7 and), and its mean value grows as $\frac{T}{2}(\log \frac{T}{2\pi} - 1) + \gamma T + o(T)$ (Figure 4), the numerical content of the hidden mean-value theorem.

Density (). Zero count $N(T) \approx F(T) = \frac{T}{2\pi}(\log \frac{T}{2\pi} - 1)$ (Definition 18, Table 1, Figures 5 and 6).

Counterpositive (). *External input, supplied by Hardy's theorem (Hardy, 1914)* : if the alignment events were finite, the energy would be bounded.

Conclusion (). The assembly map combines the five fields :

$$\text{position} \wedge \text{frequency} \wedge \text{energy} \wedge \text{density} \wedge \text{counterpositive} \implies \\ \text{infinitely many alignment events on the critical circle.}$$

The conclusion is therefore : *infinitely many zeros lie on the critical circle* — their position is the unique fixed circle $\sqrt{e}$, and their abundance is governed by the frequency-phase projection $N(T) \approx F(T)$.

10. Honest boundaries and conclusion

We catalogue precisely what is proved, what is conjectured, and what is supplied.

Formalized (Lean, no). The coordinate identities of Section 3 ; the energy balance criterion and the radius uniqueness chain of Section 4 ; the intrinsic zero-free theorem $\zeta(s) \neq 0$ for $\Re s \geq 2$ (Section 5) ; the reflection pair decomposition $\zeta(1-s) = 0 \iff \zeta(s) = 0$ with self-pairs at $\pm\sqrt{e}$ (Section 6) ; the frequency structure — gaps vanishing and antitone, phase freezing, fold point existence (Section 8) ; and the algebraic bridges between them.

Declared structures with open fields. (), (), (), ().

Numerically supported. The mean-value growth

$$E(T) = \frac{T}{2} \left(\log \frac{T}{2\pi} - 1 \right) + \gamma T + o(T)$$

(Figure 4) ; the density projection $N(T)/F(T) \rightarrow 1$ (Table 1, Figures 5–6).

Externally supplied. Hardy's bounded-energy counterpositive (), the only analytic input of the assembled argument ; it is named and imported, not proved here.

Not proved. The absence of zeros in the annulus $1 < |w| < e$ away from the critical circle — the statement of Definition 17. Within the hidden-number coordinates this is the Riemann hypothesis itself. The full strength of the present framework is the reduction : *the hypothesis is equivalent to a single statement about inversion pairs, and everything else — position, frequency, energy, density — is assembled and formally verified.*

Conclusion. The hidden-number coordinate system $w = e^s$ reduces the Riemann hypothesis to a geometric statement about the unique inversion-fixed circle $|w| = \sqrt{e}$. The formalized corpus proves the position (energy balance, radius uniqueness, reflection pairs) and the frequency and energy mechanisms; the assembled argument establishes infinitely many zeros on the critical circle, with position and with abundance, up to the single explicitly named external input. The final step — every inversion pair is self-paired, i.e. no zero off the critical circle — remains the hypothesis itself, now isolated as the unique open statement of the framework.

All formalized results were developed in the interactive theorem prover Lean (de Moura et al., 2015) against its mathematical library (The mathlib Community, 2020).

Références

- Bombieri, E. and Hejhal, D. A. (1995). On the distribution of zeros of linear combinations of Euler products. *Duke Mathematical Journal*, 80(3) :821–862.
- Connes, A. (1999). Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Mathematica, New Series*, 5(1) :29–106.
- Conrey, J. B. (1989). More than two fifths of the zeros of the Riemann zeta function are on the critical line. *Journal für die reine und angewandte Mathematik*, 399 :1–26.
- Conrey, J. B. (2003). The Riemann hypothesis. *Notices of the American Mathematical Society*, 50(3) :341–353.
- de la Vallée Poussin, C.-J. (1899). Sur la fonction Zeta(s) de Riemann et le nombre des nombres premiers inférieurs à une limite donnée. *Mémoires Couronnés et Mémoires des Savants Étrangers*, 59 :1–74.
- de Moura, L., Kong, S., Avigad, J., van Doorn, F., and von Raumer, J. (2015). The lean theorem prover (system description). In *Automated Deduction — CADE-25*, pages 378–388. Springer.
- Edwards, H. M. (1974). *Riemann’s Zeta Function*. Academic Press, New York.
- Hadamard, J. (1893). Étude sur les propriétés des fonctions entières et en particulier d’une fonction considérée par Riemann. *Journal de Mathématiques Pures et Appliquées*, 9 :171–215.
- Hardy, G. H. (1914). Sur les zéros de la fonction zeta(s) de Riemann. *Comptes Rendus de l’Académie des Sciences*, 158 :1012–1014.
- Hardy, G. H. and Littlewood, J. E. (1921). The zeros of Riemann’s zeta-function on the critical line. *Mathematische Zeitschrift*, 10(3–4) :283–317.
- Heath-Brown, D. R. (1979). Simple zeros of the Riemann zeta-function on the critical line. *Bulletin of the London Mathematical Society*, 11 :17–18.
- Iwaniec, H. (2014). *Lectures on the Riemann Zeta Function*, volume 62 of *University Lecture Series*. American Mathematical Society, Providence, RI.
- Iwaniec, H. and Kowalski, E. (2004). *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI.
- Lagarias, J. C. (2002). An elementary problem equivalent to the Riemann hypothesis. *The American Mathematical Monthly*, 109(6) :534–543.
- Levinson, N. (1974). More than one third of zeros of Riemann’s zeta-function are on $\sigma = 1/2$. *Advances in Mathematics*, 13 :383–436.
- Liu, Y.-J. and Xu, Y.-N. (2026). The hidden-space bridge system : A three-axiom foundation connecting the real and imaginary domains. *aiXiv preprint no. aioxiv.260826.000001*.
- Montgomery, H. L. (1973). The pair correlation of zeros of the zeta-function. In *Proceedings of Symposia in Pure Mathematics*, volume 24, pages 181–193, Providence, RI. American Mathematical Society.
- Montgomery, H. L. (1977). Extreme values of the Riemann zeta-function. *Commentarii Mathematici Helvetici*, 52 :511–518.
- Platt, D. J. and Trudgian, T. (2021). The Riemann hypothesis is true up to 3×10^{12} . *Bulletin of the London Mathematical Society*, 53(3) :792–797.
- Riemann, B. (1859). Über die anzahl der primzahlen unter einer gegebenen größe. *Monatsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin*, pages 671–680.
- Schoenfeld, L. (1976). Sharper bounds for the chebyshev functions

- $\theta(x)$ and
 $\psi(x)$, ii. *Mathematics of Computation*, 30 :337–360.
- Selberg, A. (1942). On the zeros of Riemann’s zeta-function on the critical line. *Skrifter utgitt av Det Norske Videnskaps-Akademi i Oslo, I. Matematisk-Naturvidenskapelig Klasse, no. 10*, pages 1–59.
- The mathlib Community (2020). The lean mathematical library. arXiv :1910.11832.
- Titchmarsh, E. C. (1986). *The Theory of the Riemann Zeta-Function*. Clarendon Press, Oxford, 2nd edition. Revised by D. R. Heath-Brown.
- von Koch, H. (1901). Sur la distribution des nombres premiers. *Acta Mathematica*, 24 :159–182.
- von Mangoldt, H. (1905). Zur verteilung der nullstellen der riemannschen funktion. *Mathematische Annalen*, 61 :1–19.